

CS & IT ENGINEERING



'C' PROGRAMMING

Practice Session

Lecture No.- 03



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Topics to be Covered



- Practice session



#Q. Consider the following C declaration
struct {

short s[5]; = $5 \times 2 = 10 \text{ Bytes}$

union {

float y; = 4 Bytes

long z; = 8 Bytes

} u; = $\text{Max}(4, 8) = 8 \text{ Bytes}$

}t; = $\text{Sum}(10 + 8) = 18 \text{ Bytes}$

Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t, ignoring alignment considerations, is

- A. 22 Bytes
- B. 14 Bytes
- ☒ C. 18 Bytes
- D. 10 Bytes



#Q. The following C declarations

```
struct node{
```

```
    int i;
```

```
    float j;
```

```
};
```

```
struct node *s[10];
```

define s to be

$(*s)[10] \Rightarrow$ Pointer to array of 10 Elements

$*s[10] \Rightarrow$ An array of 10 Pointers

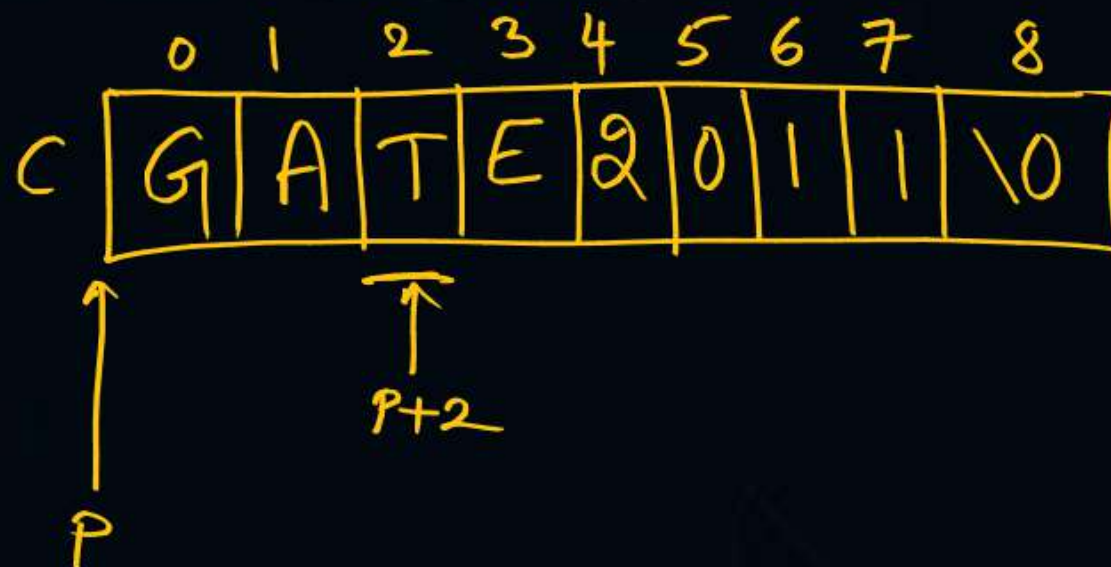
- ☒ A. An array, each element of which is a pointer to a structure of type node
- ☒ B. A structure of 2 fields, each field being a pointer to an array of 10 elements
- ☒ C. A structure of 3 fields: an integer, a float, and an array of 10 elements
- ☐ D. An array, each element of which is a structure of type node



#Q. What does the following fragment of C-program print?

```
char c[] = "GATE2011";  
char *p = c;  
printf("%s", p + p[0] - p[3]);
```

- (A) GATE2011
- (B) ATE2011
- ✓ (C) TE2011
- (D) 2011



$p + 'G' - 'E'$

$p + 2$



#Q. What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.

```
#include <stdio.h>
```

```
int main()
```

```
{  
    unsigned int x[4][3] = {{1, 2, 3}, {4, 5, 6},  
                           {7, 8, 9}, {10, 11, 12}};
```

```
    printf("%u, %u, %u", x+3, *(x+3), *(x+2)+3);  
}
```

☒ A. 2036, 2036, 2036

☒ B. 2012, 4, 2204

☒ C. 2036, 10, 10

☒ D. 2012, 4, 6

	0	1	2
x	2000	2004	2008
x+0 *(x+0) →	1	2	3
x+1 *(x+1) →	4	5	6
x+2 *(x+2) →	7	8	9
x+3 *(x+3) →	10	11	12

$$2024 + 3 \times 4 = 2024 + 12 = 2036$$



Topic : Practice Session



#Q. What will be the output of the C program, if executed on a 32-bit processor?

```
#include<stdio.h>
```

```
void function(char**);
```

```
int main()
```

```
{    char *arr[] = { "ant", "bat", "cat", "dog", "egg", "fly" };
```

```
    function(arr);
```

```
    return 0;
```

```
void function(char **ptr)
```

```
    char *ptr1;
```

```
    ptr1 = (ptr += sizeof(int))[-2];
```

```
    printf("%s\n", ptr1);
```

A. bat

☒ B. cat

C. ant

D. dog

$\Rightarrow ptr+2$

Diagram illustrating the memory layout and pointer manipulation:

Array *arr* is stored in memory, with each string occupying 4 bytes (32-bit processor):

a	n	t	\0
b	a	t	\0
c	a	t	\0
d	o	g	\0
e	g	g	\0
f	l	y	\0

Handwritten annotations show pointer arithmetic:

- $ptr = ptr + 4$ (moving to the start of the second string, "bat")
- $(ptr)[-2]$ (accessing the character at index -2, which is 'c' from the third string, "cat")
- $ptr1 = (ptr + 4)[-2]$ (storing 'c' in *ptr1*)



#Q. What is printed by the print statements in the program assuming call by reference parameter passing?

Program P1() {

x = 10;

y = 3;

func1(y, x, x);

print x;

print y; }

func1(x, y, z) {

* y = *y + 4;

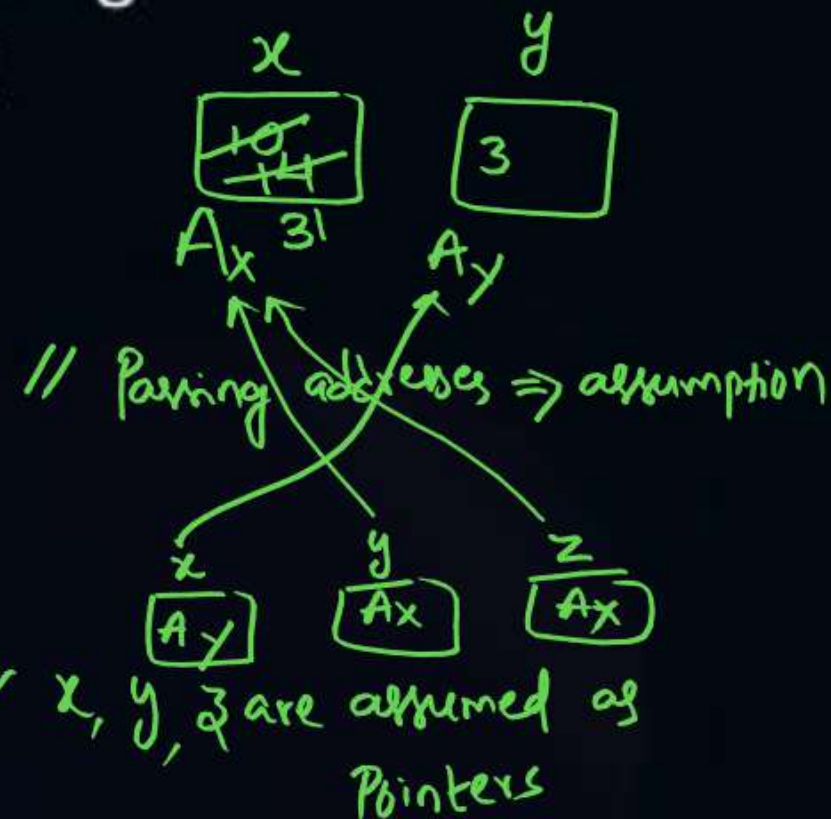
*z = *x + *y + *z; }

A. 10, 3

☒ B. 31, 3

C. 27, 7

D. None of the above



when arguments are Pointers,
Then while calling address is Passed.
(Reference)



#Q. The following program fragment is written in a programming language that allows global variables and does not allow nested declarations of functions.

```
global int i=100, j=205;  
void P(x) { x=105  
int i=10; 200  
print(x+10); //115  
i=200;  
j=20;  
print (x); //105  
}
```

- A. 115, 220
- B. 25, 220
- C. 25, 15
- ☒ D. 115, 105

→ main() {P(i+j);} ^{P(105);}

If the programming language uses static scoping and call by ^{value} need parameter passing mechanism, the values printed by the above program are:



Topic : Practice Session – GATE 2008



#Q. Consider the C program below. What does it print?

```
# include <stdio.h>
```

```
# define swap1 (a, b) tmp = a; a = b; b = tmp
```

```
void swap2 ( int a, int b){
```

```
int tmp;
```

```
tmp = a; a = b; b = tmp;}
```

```
✓ void swap3 (int*a, int*b){
```

```
int tmp;
```

```
tmp = *a; *a = *b; *b = tmp;}
```

```
→ int main (){
```

```
int num1 = 5, num2 = 4, tmp;
```

```
if (num1 < num2) {swap1 (num1, num2);}
```

```
if (num1 < num2) {swap2 (num1 + 1, num2);}
```

```
if (num1 >= num2) {swap3 (&num1, &num2);}
```

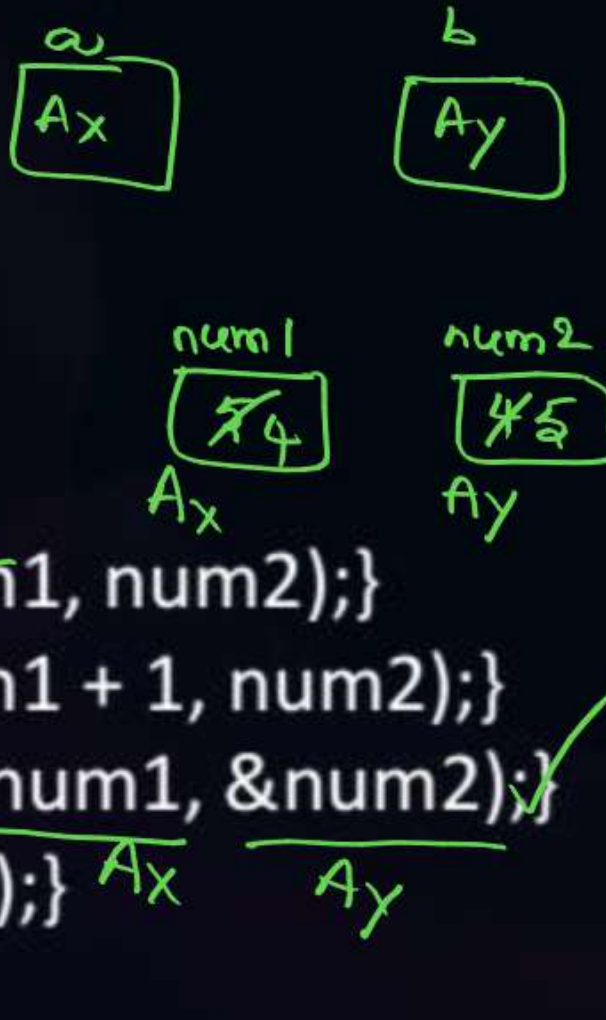
```
printf ("%d, %d", num1, num2);}
```

(A) 5, 5

(B) 5, 4

✓ (C) 4, 5

(D) 4, 4





Topic : Practice Session – GATE 2006

→, Postfix same Precedence L To R

Prefix less Precedence R To L



#Q. Which one of the choices given below would be printed when the following program is executed?

Let 1 int = 4 Bytes

```
struct test {
```

```
    int i;    = 4 Bytes
```

```
    char *c;  = 4 Bytes
```

```
}st[] = {5, \"become\", 4, \"better\", 6, \"jungle\", 8, \"ancestor\", 7, \"brother\"};
```

```
main () {
```

```
    struct test *p = st; ✓
```

```
    p += 1; ✓
```

```
    ++p -> c; ++(p->c) ✓
```

```
    printf(\"%s\", p++ -> c); ✓
```

```
    printf(\"%c\", *++p -> c);
```

```
    printf(\"%d\", p[0].i); // 6
```

```
    printf(\"%s \\n\", p -> c); } // Jungle
```

~~(A)~~ jungle, n, 8, nclastor

~~(B)~~ etter, u, 6, ungle

(C) cetter, k, 6, jungle

~~(D)~~ etter, u, 8, ncestor

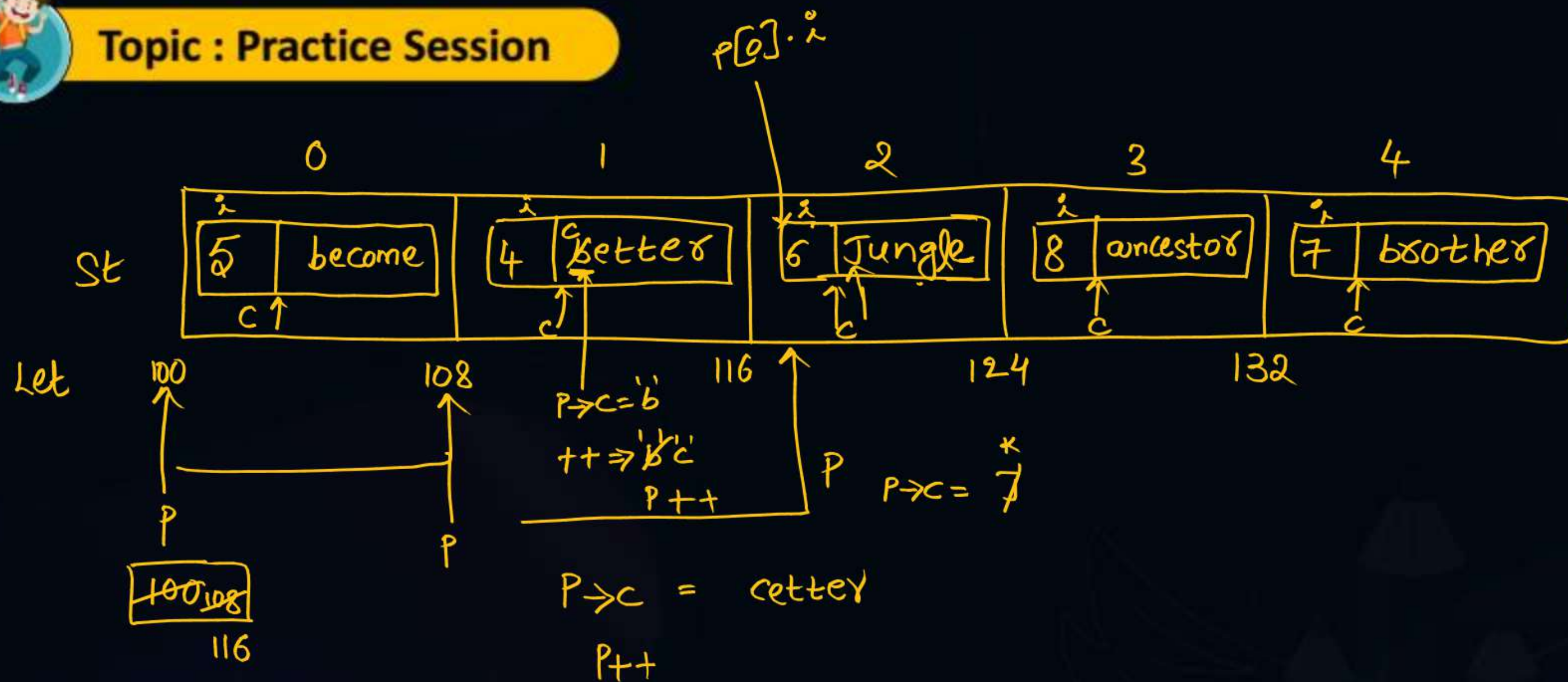
++ (Postfix)

→ (p->c) then p++ // cetter

p->c, ++, * // k



Topic : Practice Session





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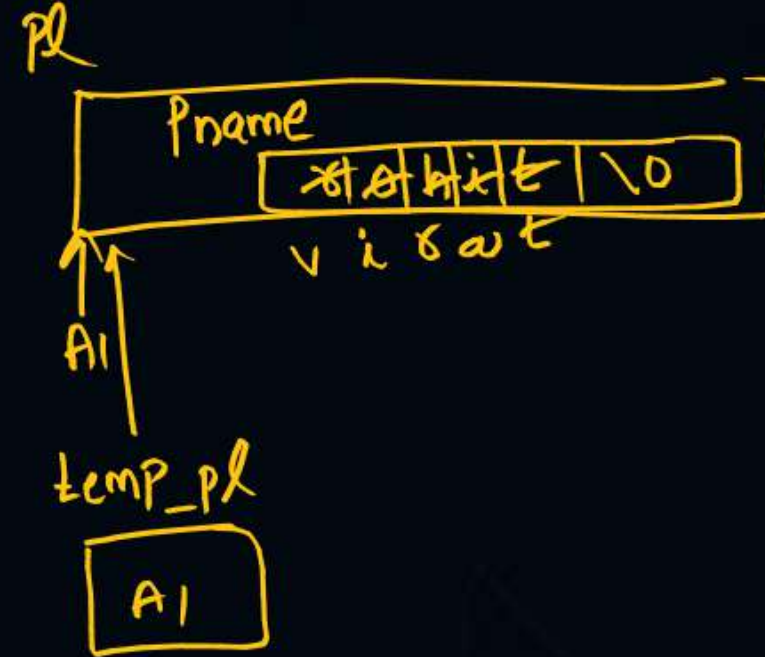


#Q. What will be the output of the C program?

```
struct player {  
    char pname[20];  
}pl;  
char* play(struct player *temp_pl){  
    strcpy(temp_pl->pname, "virat");  
    return temp_pl->pname;  
}
```

```
→int main(){  
    strcpy(pl.pname, "rohit");  
    printf("%s %s", pl.pname, play(&pl));  
    return 0; }
```

① ~~virat~~
② ~~virat~~
← Evaluates from Right to Left



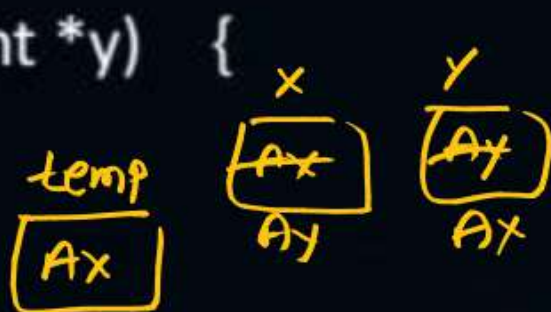
- (A) rohit virat
- ☒ (B) virat virat
- (C) rohit rohit
- (D) virat rohit



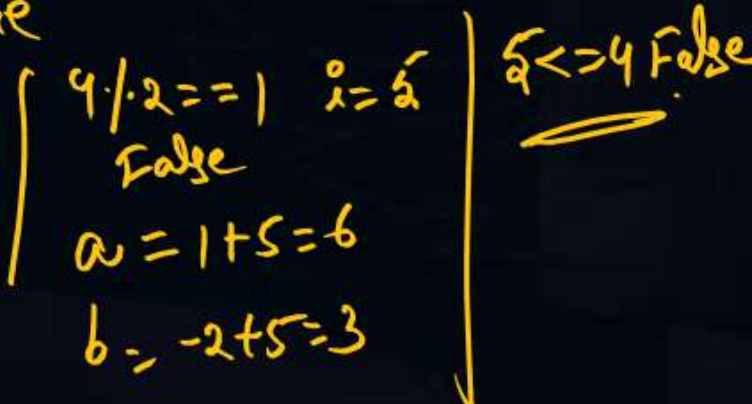
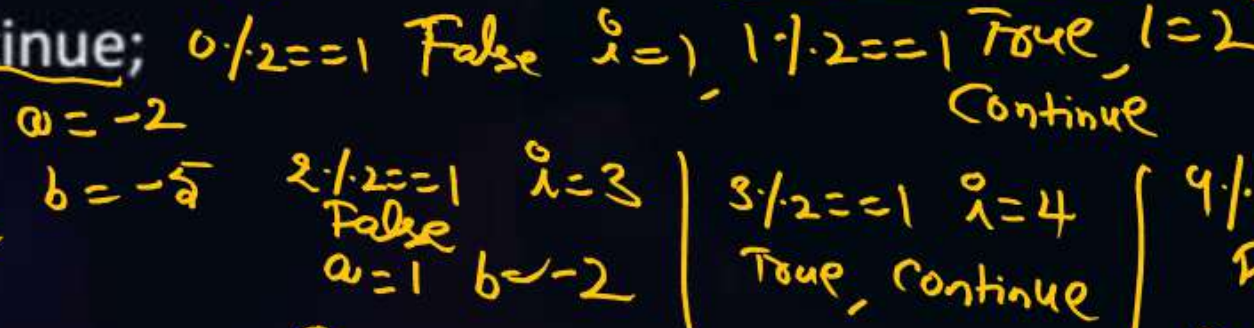
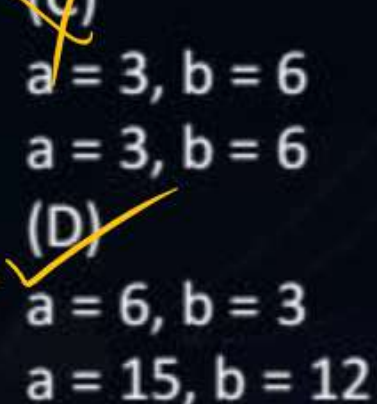
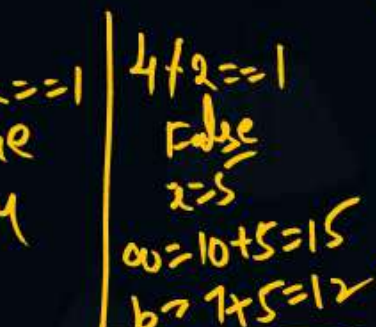
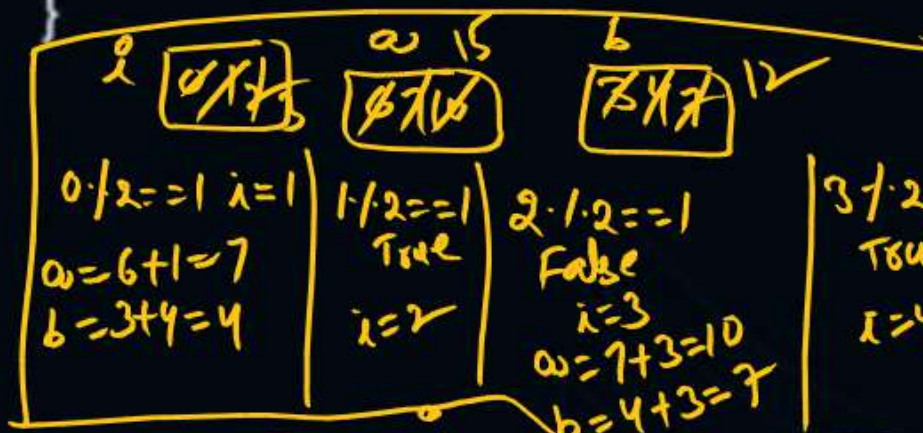
#Q. Which one of the choices given below would be printed when the following program is executed?

```
void swap (int *x, int *y) {  
    static int *temp;  
    temp = x;  
    x = y;  
    y = temp;  
}
```

```
void printab ()  
{  
    static int i, a = -3, b = -6;  
    i = 0;  
    while (i <= 4){  
        if ((i++)%2 == 1) continue;  
        a = a + i;  
        b = b + i;  
        swap (&a, &b);  
        printf("a = %d, b = %d\n", a, b);  
    }  
}
```



```
main()  
{  
    printab();  
    printab();  
}
```



6 3
15 12

- (A) a = 0, b = 3
- (B) a = 3, b = 0
- (C) a = 12, b = 9
- (D) a = 3, b = 6



Topic : Practice Session – GATE 2012



#Q. Consider the following C program

```
int a, b, c = 0;
void prtFun (void);
→ int main ()
{
    static int a = 1;
    prtFun();
    a += 1;
    prtFun();
    printf ( "\n %d %d " , a, b) ;
}
```

```
void prtFun (void)
```

```
{
    static int a = 2;
    int b = 1;
    a += ++b;
    printf ( " \n %d %d " , a, b);
}
```

a. 3 1

4 1

4 2

b. 4 2

6 1

6 1

✓ c. 4 2

6 2

2 0

d. 3 1

5 2

5 2

What output will be generated by the given code segment?



Topic : Practice Session – GATE 2012



#Q. Consider the following C program

```
int a, b, c = 0;
void prtFun (void);
int main ()
{
    → auto int a = 1;
    prtFun();
    a += 1;
    prtFun();
    printf ( "\n %d %d " , a, b) ;
}
```

void prtFun (void)

```
{
    → register int a = 2;
    int b = 1;
    a += ++b;
    printf ( " \n %d %d " , a, b);
}
```

$$a = 2 + 2 = 4$$

$$a = 2 + 2 = 4$$

4 2
4 2

o/p= _____

4 2

4 2

2 0

a. 3 1

4 1

4 2

b. 4 2

6 1

6 1

c. 4 2

6 2

2 0

d. 3 1

5 2

5 2

What output will be generated by the given code segment?



#Q. Consider the following snippet of a C program. Assume that swap (&x, &y) exchanges the content of x and y : 1>2

→ int main () {
 int array[] = {~~3~~, ~~5~~, ~~1~~, ~~4~~, ~~6~~, ~~2~~};
 int done = 0;
 int i;
 while (done == 0) {
 done = 1; ✓
 for (i = 0; i <= 4; i++) {
 if (array[i] < array[i+1]) {
 swap(&array[i], &array[i+1]);
 done = 0;
 }
 }
 }

Handwritten notes for the first loop:
- Above array indices 0-5: 0, 1, 2, 3, 4, 5
- Above array values: 6, 5, 4, 3, 2, 1 (with arrows indicating swaps between adjacent elements)
- To the right of the if condition: 3 < 5 True, 5 < 4 False, 4 < 3 True, 3 < 2 True, 2 < 1 True
- Below the for loop: } // End of if, } // End of for

array

0	1	2	3	4	5
6	5	4	3	2	1

=

for (i = 5; i >= 1; i--) {
 if (array[i] > array[i-1]) {
 swap(&array[i], &array[i-1]);
 done = 0;
 } // End of if
} // End of for
} // End of while
printf("%d", array[3]);
}

Handwritten notes for the second loop:
- To the left of the if condition: False always
- To the right of the loop: This loop has No impact
- Below the printf statement: The output of the program is 3



#Q. For the program given below what will be printed by the write statements marked and in the program if the variables are Statically scoped?

```
var x, y:integer;
```

```
procedure P(n:integer);
```

```
begin
```

```
x := (n+2)/(n-3);
```

```
end;
```

```
procedure Q
```

```
var x, y:integer;
```

```
begin
```

```
x:=3;
```

```
y:=4;
```

```
P(y);
```

```
Write(x); __ (1)
```

```
end;
```

```
begin
```

```
x:=7;
```

```
y:=8;
```

```
Q;
```

```
Write(x); __ (2)
```

```
end.
```

A. 3, 6

B. 6, 7

C. 3, 7

D. None of the above



#Q. there is an ^X in the program which indicates some missing statements. Choose the correct option to replace in the program. *To Compute Transpose given matrix.*

```
#include<stdio.h>
```

```
#define ROW 4
```

```
#define COL 4
```

```
int M[ROW][COL] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16};
```

```
main(){
```

```
int i, j, t;
```

```
for (i = 0; i < 4; ++i)
```

```
{
```

```
X //missing Part of Code
```

```
}
```

```
for (i = 0; i < 4; ++i)
```

```
for (j = 0; j < 4; ++j)
```

```
printf ("%d", M[i][j]);
```

```
}
```



Topic : Practice Session



```
A. for(j = 0; j < 4; ++j){  
  t = M[i][j];  
  M[i][j] = M[j][i];  
  M[j][i] = t;  
}
```

```
B. for(j = 0; j < 4; ++j){  
  M[i][j] = t;  
  t = M[j][i];  
  M[j][i] = M[i][j];  
}
```

```
C. for(j = i; j < 4; ++j){  
  t = M[i][j];  
  M[i][j] = M[j][i];  
  M[j][i] = t;  
}
```

```
D. for(j = i; j < 4; ++j){  
  M[i][j] = t;  
  t = M[j][i];  
  M[j][i] = M[i][j];  
}
```




THANK - YOU